

INTRODUCTION TO DATA SCIENCE

EDA

Data analysis for the Titanic

The questions we are going to solve are:

1. *Was the famous phrase “Women and kids first” said?*
2. *In which port embarked most part of rich people?*
3. *Where did most people survive? Does it coincide with where richer people embarked?*
4. *Can money save lives?*
5. *What is the relationship between the age and the type of class people embarked on?*

1. Was the famous phrase “Women and kids first” said?

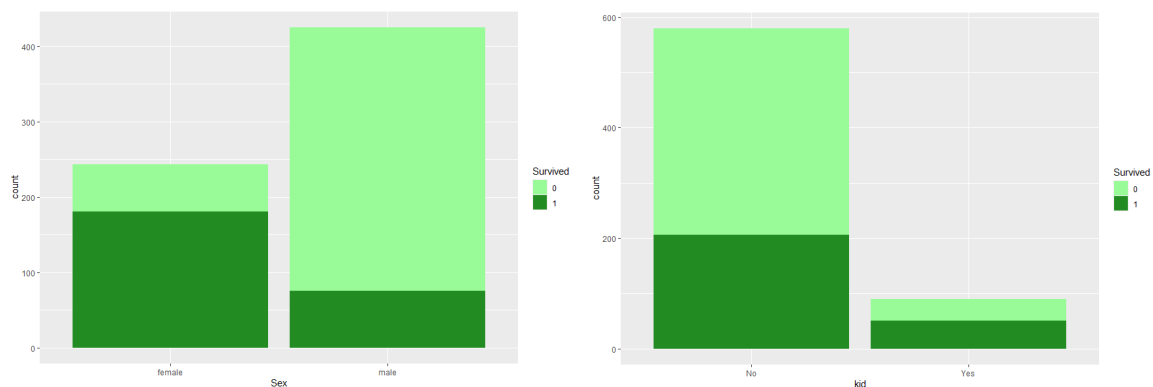
It may be the most common question because the solution is quite logical. On those years, male individuals were taught to protect the females and treat them with respect in fact, when there was any type of emergency where people had to get out of somewhere (a building, a plane, a cave, a boat...) it was usually said: “Women and kids first”. It seems that on the titanic the same sentence was said and followed as an order because of the data we can see on the following table of values:

	0	1
female	0.2551440	0.7448560
male	0.8235294	0.1764706

In this table, where 0 means the ones who did not survive and 1 the ones who did, we can clearly observe that nearly 3 of each 4 women survived to the titanic, while in case of men only 1 of each 5 had the possibility to save his own life. Men could only increase their chances of surviving if they were under 18 years, this would increase their probabilities up to 56% as we can see on this other table:

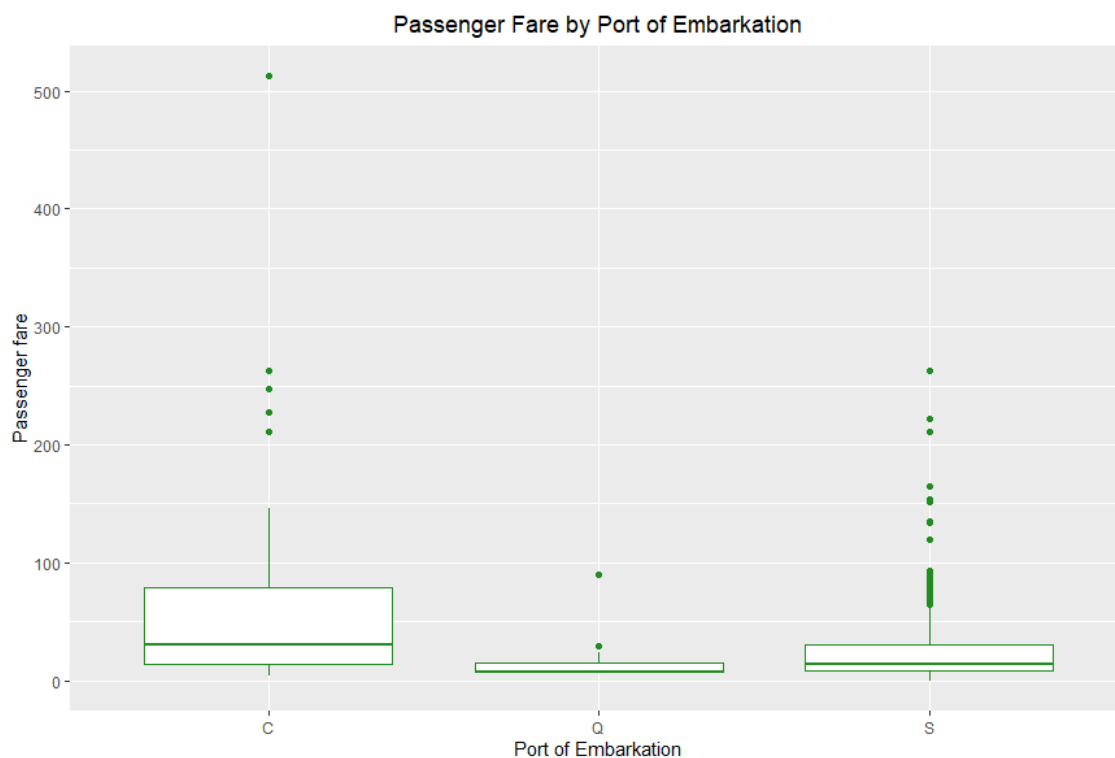
boys	0	1
No	0.6442142	0.3557858
Yes	0.4382022	0.5617978

After considering all this data we can affirm that the famous phrase was obeyed. Below you have some plots that show the previous information more graphically.



2. In which port embarked most part of rich people?

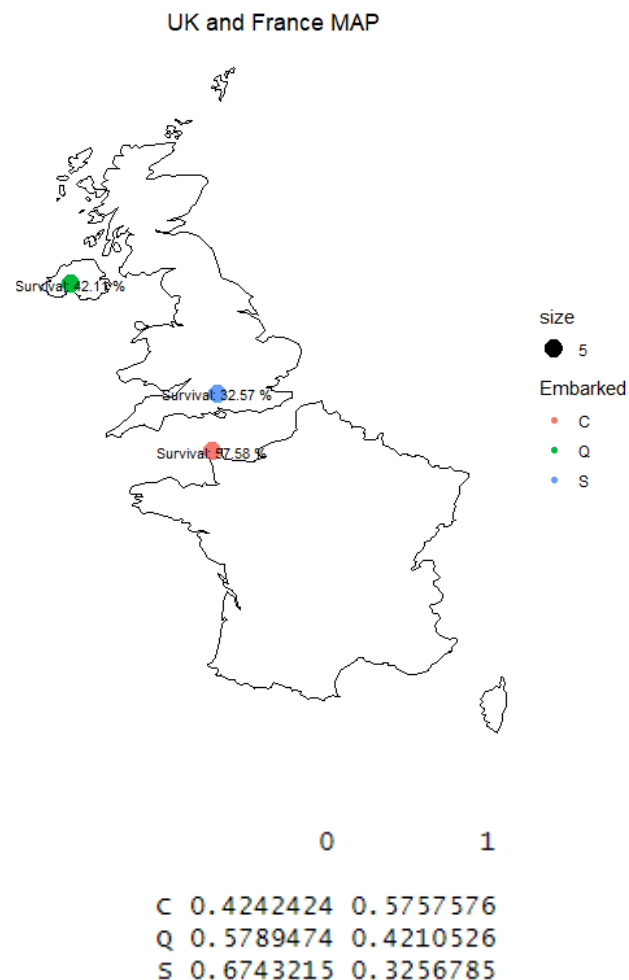
Comparing the data given, we resume it in the following boxplot:



In this boxplot we can see that the port of embarkation affects the passenger's fare. The highest fare is in Cherbourg, as well as the richest people. Next, we have Southampton, the curiosity of this case is observed on the great quantity of outliers. Finally, is Queenstown that has very low fares and a few outliers.

3. Where did most people survive? Does it coincide with where more rich people embarked?

We must observe the data of how many people survived of each port. We can visualize this information in a clearer way if we see it in a map like the following one:

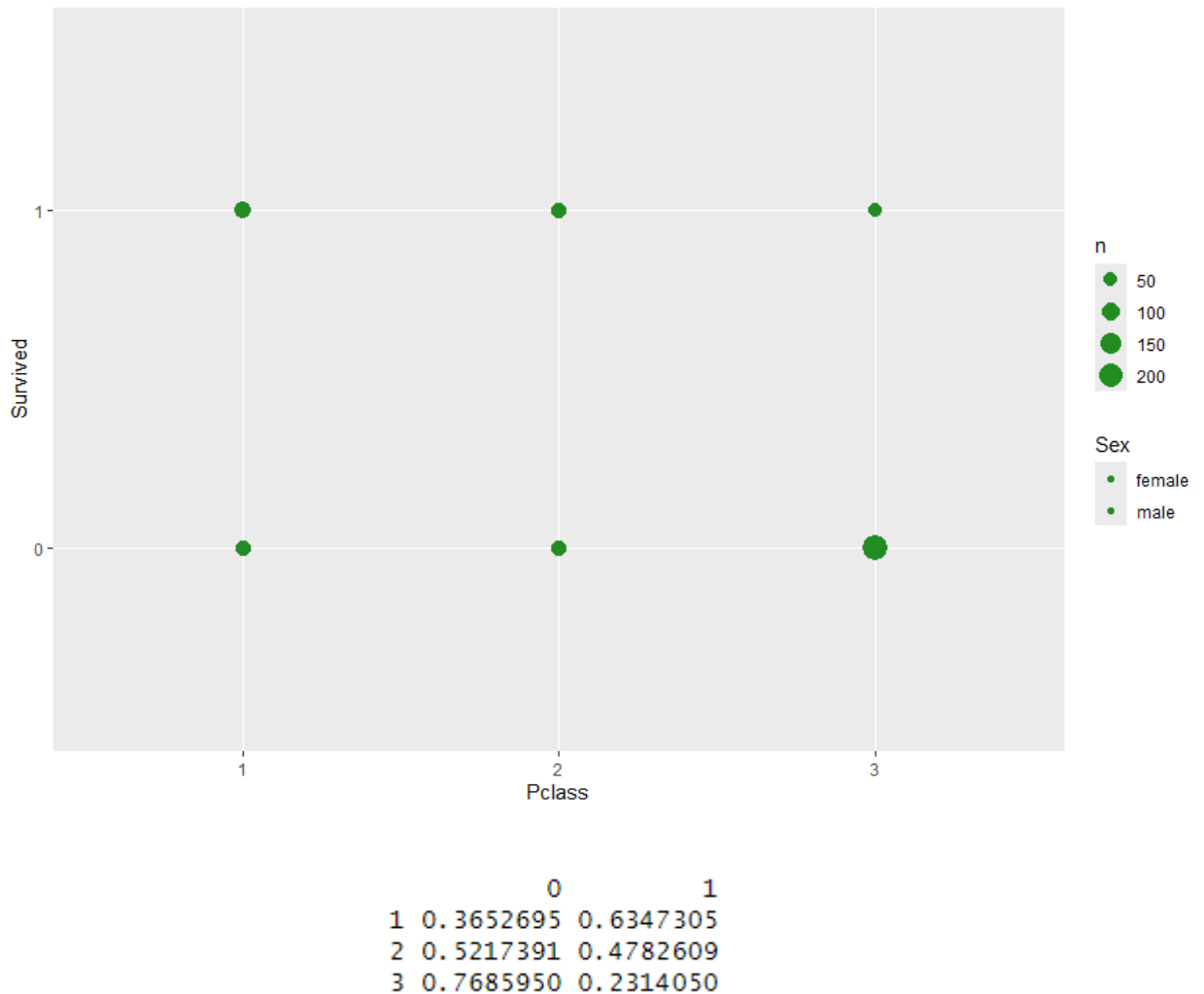


Here we are watching the map of UK and France in order to see the ports of embarkation: Cherbourg, Queenstown, Southampton. Also, there is the percentage of survivors of each port, being the lowest in Southampton with 32.57%, followed by Queenstown with a 42.11% and at the top Cherbourg with a 57.58%. We got these values from the table below the map.

We can observe it coincides that Cherbourg has the highest percentage of survival rate (57.58%) and the most amount of rich people embarked. Queenstown is the second port with a higher survival rate than Southampton, even though in this last one embarked more rich people. This lets us deduce that the survival chances do not depend much on in which port one embarked.

4. Can money save lives?

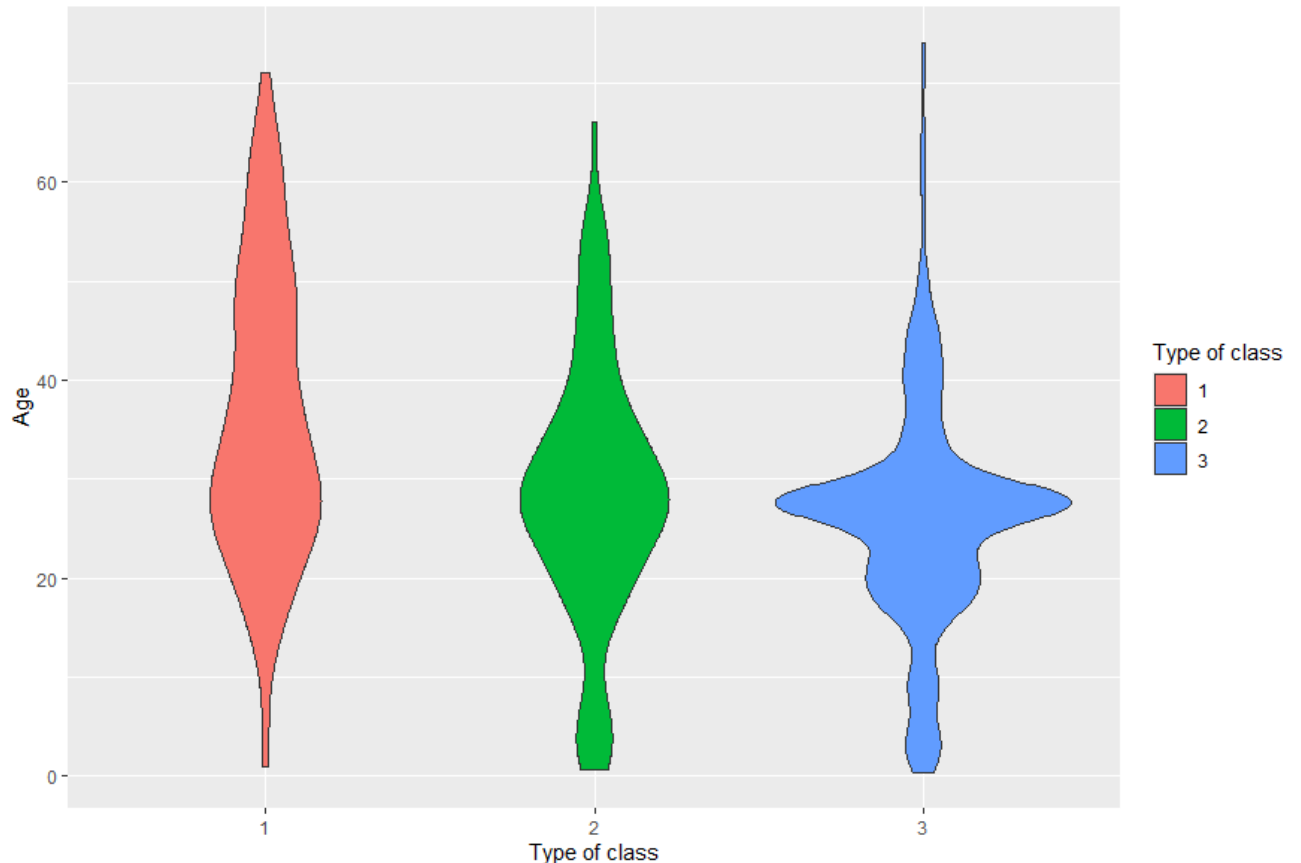
Money has always been an important factor on people's lives. The ones who travelled in first class used to be rich people who, on those dates, were more important than normal people because the fact that the amount of money that one had determined its social status and its importance on society. Below we can see some data which will help us to solve the question.



We can observe that most people were in third class, what is quite logical because on those days most of the population did not have a high fortune. We can then deduce that the better class you were the more probability of surviving you had, which confirms that money in fact can save lives.

5. What is the relation between the age and the type of class people embarked on?

Summing up all the information we have of the variables related to the questions we end on this violin plot from which we are going to have a more visual image of the data:



We can visualize that age really affects the type of class of the people. The oldest people tend to be first class while people between 15-45 years are more in second and third class. Also, we can see that people between 25-30 are mostly in third class. To end, kids tend to be second and especially third class.

Conclusion

With the assignment we can observe the following data: women and children were given priority in the lifeboats during the sinking of the Titanic; secondly, most of the rich passengers boarded at Cherbourg, followed by Southampton and Queenstown, although this has no great relevance in terms of the survival of the passengers, as we have been able to see on the map; moreover, first class passengers had a higher survival rate compared to second and third class passengers, this could be due to the location of the cabins on the ship or other considerations such as the social status; last but not least, and regarding the age distribution, we can see that the majority of the passengers were young and middle-aged adults. Furthermore, you can see that the majority of seniors tend to be in the first class, while the middle-aged passengers are more willing to be at second and especially third class.